# IPv6 Address Types

12.3.1

## Unicast, Multicast, Anycast

As with IPv4, there are different types of IPv6 addresses. In fact, there are three broad categories of IPv6 addresses:

- **Unicast** - An IPv6 unicast address uniquely identifies an interface on an IPv6-enabled device.
- **Multicast** - An IPv6 multicast address is used to send a single IPv6 packet to multiple destinations.
- **Anycast** - An IPv6 anycast address is any IPv6 unicast address that can be assigned to multiple devices. A packet sent to an anycast address is routed to the nearest device having that address. Anycast addresses are beyond the scope of this course.

Unlike IPv4, IPv6 does not have a broadcast address. However, there is an IPv6 all-nodes multicast address that essentially gives the same result.

12.3.2

## IPv6 Prefix Length

The prefix, or network portion, of an IPv6 address can be identified by a dotted-decimal subnet mask or prefix length (slash notation). For example, an IPv4 address of 192.168.1.10 with dotted-decimal subnet mask 255.255.255.0 is equivalent to 192.168.1.10/24.

In IPv6 it is only called the prefix length. IPv6 does not use the dotted-decimal subnet mask notation. Like IPv4, the prefix length is represented in slash notation and is used to indicate the network portion of an IPv6 address.

The prefix length can range from 0 to 128. The recommended IPv6 prefix length for LANs and most other types of networks is /64, as shown in the figure.

### IPv6 Prefix Length

The prefix or network portion of the address is 64 bits in length, leaving another 64 bits for the interface ID (host portion) of the address.

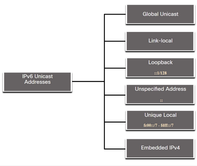
It is strongly recommended to use a 64-bit Interface ID for most networks. This is because stateless address autoconfiguration (SLAAC) uses 64 bits for the Interface ID. It also makes subnetting easier to create and manage.

12.3.3

## Types of IPv6 Unicast Addresses

An IPv6 unicast address uniquely identifies an interface on an IPv6-enabled device. A packet sent to a unicast address is received by the interface which is assigned that address. Similar to IPv4, a source IPv6 address must be a unicast address. The destination IPv6 address can be either a unicast or a multicast address. The figure shows the different types of IPv6 unicast addresses.

### IPv6 Unicast Addresses




Unlike IPv4 devices that have only a single address, IPv6 addresses typically have two unicast addresses:

- **Global Unicast Address (GUA)** - This is similar to a public IPv4 address. These are globally unique, internet-routable addresses. GUAs can be configured statically or assigned dynamically.
- **Link-local Address (LLA)** - This is required for every IPv6-enabled device. LLAs are used to communicate with other devices on the same local link. With IPv6, the term link refers to a subnet. LLAs are confined to a single link. Their uniqueness must only be confirmed on that link because they are not routable beyond the link. In other words, routers will not forward packets with a link-local source or destination address.

12.3.4

## A Note About the Unique Local Address

Unique local addresses (range fc00::/7 to fdff::/7) are not yet commonly implemented. Therefore, this module only covers GUA and LLA configuration. However, unique local addresses may eventually be used to address devices that should not be accessible from the outside, such as internal servers and printers.

The IPv6 unique local addresses have some similarity to RFC 1918 private addresses for IPv4, but there are significant differences:

- Unique local addresses are used for local addressing within a site or between a limited number of sites.
- Unique local addresses can be used for devices that will never need to access another network.
- Unique local addresses are not globally routed or translated to a global IPv6 address.

**Note**: Many sites also use the private nature of RFC 1918 addresses to attempt to secure or hide their network from potential security risks. However, this was never the intended use of these technologies, and the IETF has always recommended that sites take the proper security precautions on their internet-facing router.

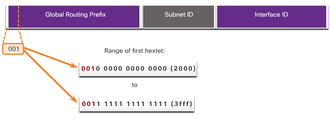
12.3.5

## IPv6 GUA

IPv6 global unicast addresses (GUAs) are globally unique and routable on the IPv6 Internet. These addresses are equivalent to public IPv4 addresses. The Internet Committee for Assigned Names and Numbers (ICANN), the operator for IANA, allocates IPv6 address blocks to the five RIRs. Currently, only GUAs with the first three bits of 001 or 2000::/3 are being assigned, as shown in the figure.

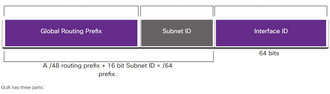
The figure shows the range of values for the first hextet where the first hexadecimal digit for currently available GUAs begins with a 2 or a 3. This is only 1/8th of the total available IPv6 address space, excluding only a very small portion for other types of unicast and multicast addresses.

**Note**: The 2001:db8::/32 address has been reserved for documentation purposes, including use in examples.




The next figure shows the structure and range of a GUA.

### IPv6 Address with a /48 Global Routing Prefix and /64 Prefix




A GUA has three parts:

- Global Routing Prefix
- Subnet ID
- Interface ID

12.3.6

## IPv6 GUA Structure

**Global Routing Prefix**

The global routing prefix is the prefix, or network, portion of the address that is assigned by the provider, such as an ISP, to a customer or site. For example, it is common for ISPs to assign a /48 global routing prefix to its customers. The global routing prefix will usually vary depending on the policies of the ISP.

The previous figure shows a GUA using a /48 global routing prefix. /48 prefixes are a common global routing prefix that is assigned and will be used in most of the examples throughout this course.

For example, the IPv6 address 2001:db8:acad::/48 has a global routing prefix that indicates that the first 48 bits (3 hextets) (2001:db8:acad) is how the ISP knows of this prefix (network). The double colon (::) following the /48 prefix length means the rest of the address contains all 0s. The size of the global routing prefix determines the size of the subnet ID.

**Subnet ID**

The Subnet ID field is the area between the Global Routing Prefix and the Interface ID. Unlike IPv4 where you must borrow bits from the host portion to create subnets, IPv6 was designed with subnetting in mind. The Subnet ID is used by an organization to identify subnets within its site. The larger the subnet ID, the more subnets available.

**Note**: Many organizations are receiving a /32 global routing prefix. Using the recommended /64 prefix in order to create a 64-bit Interface ID, leaves a 32 bit Subnet ID. This means an organization with a /32 global routing prefix and a 32-bit Subnet ID will have 4.3 billion subnets, each with 18 quintillion devices per subnet. That is as many subnets as there are public IPv4 addresses!

The IPv6 address in the previous figure has a /48 Global Routing Prefix, which is common among many enterprise networks. This makes it especially easy to examine the different parts of the address. Using a typical /64 prefix length, the first four hextets are for the network portion of the address, with the fourth hextet indicating the Subnet ID. The remaining four hextets are for the Interface ID.

**Interface ID**

The IPv6 interface ID is equivalent to the host portion of an IPv4 address. The term Interface ID is used because a single host may have multiple interfaces, each having one or more IPv6 addresses. The figure shows an example of the structure of an IPv6 GUA. It is strongly recommended that in most cases /64 subnets should be used, which creates a 64-bit interface ID. A 64-bit interface ID allows for 18 quintillion devices or hosts per subnet.

A /64 subnet or prefix (Global Routing Prefix + Subnet ID) leaves 64 bits for the Interface ID. This is recommended to allow SLAAC-enabled devices to create their own 64-bit Interface ID. It also makes developing an IPv6 addressing plan simple and effective.

**Note**: Unlike IPv4, in IPv6, the all-0s and all-1s host addresses can be assigned to a device. The all-1s address can be used because broadcast addresses are not used within IPv6. The all-0s address can also be used, but is reserved as a Subnet-Router anycast address, and should be assigned only to routers.

12.3.7

## IPv6 LLA

An IPv6 link-local address (LLA) enables a device to communicate with other IPv6-enabled devices on the same link and only on that link (subnet). Packets with a source or destination LLA cannot be routed beyond the link from which the packet originated.

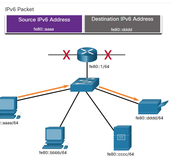
The GUA is not a requirement. However, every IPv6-enabled network interface must have an LLA.

If an LLA is not configured manually on an interface, the device will automatically create its own without communicating with a DHCP server. IPv6-enabled hosts create an IPv6 LLA even if the device has not been assigned a global unicast IPv6 address. This allows IPv6-enabled devices to communicate with other IPv6-enabled devices on the same subnet. This includes communication with the default gateway (router).

IPv6 LLAs are in the fe80::/10 range. The /10 indicates that the first 10 bits are 1111 1110 10xx xxxx. The first hextet has a range of 1111 1110 1000 **0000** (fe80) to 1111 1110 1011 **1111** (febf).

The figure shows an example of communication using IPv6 LLAs. The PC is able to communicate directly with the printer using the LLAs.

### IPv6 Link-Local Communications




The next figure shows some of the uses for IPv6 LLAs.

1. Routers use the LLA of neighbor routers to send routing updates.
2. Hosts use the LLA of a local router as the default gateway.

**Note**: Typically, it is the LLA of the router, and not the GUA, that is used as the default gateway for other devices on the link.

There are two ways that a device can obtain an LLA:

- **Statically** – This means the device has been manually configured.
- **Dynamically** – This means the device creates its own Interface ID by using randomly generated values or using the Extended Unique Identifier (EUI) method, which uses the client MAC address along with additional bits.

12.3.8

## Check Your Understanding - IPv6 Address Types

Check your understanding of IPv6 address types by choosing the correct answer to the following questions.

1. What is the recommended prefix length for most IPv6 subnets?
   - /32
   - /48
   - /64
   - /128

2. Which part of a GUA is assigned by the ISP?
   - global routing prefix
   - global routing prefix and subnet ID
   - prefix
   - /48 prefix

3. Which type of IPv6 unicast address is not routable between networks?
   - unique local address
   - GUA
   - embedded IPv4 address
   - LLA

4. True or false? The Subnet ID field in an GUA must borrow bits from the interface ID.
   - True
   - False

5. What type of IPv6 address begins with fe80?
   - GUA
   - LLA
   - multicast address
   - None. An IPv6 address must begin with 2001.

Check

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Reset